

HUANXUAN LIAO

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🎓 EDUCATION

Institute of Automation, Chinese Academy of Sciences (CASIA), Beijing 2023 – Present

PhD student in Pattern Recognition and Intelligent Systems (NLPR). Advisors: [Shizhu He](#) and [Kang Liu](#)

North China Electric Power University (NCEPU), Beijing

2019 – 2023

Bachelor of Engineering in Intelligence Science and Technology (Innovation Class). Ranked 1/13

💡 RESEARCH INTERESTS

I work toward **foundation models for long-horizon adaptive intelligence**: models that use growing contexts effectively, retain reusable capabilities, and act reliably over extended interactions.

- **Long-Context Data & Model Co-Design**: long-context data curation and synthesis, length- and capability-aware mixtures, scalable pretraining and mid-training, efficient attention, KV-cache optimization, and architectures to jointly improve long-context capability, scalability, and efficiency ([LCLM-Horizon](#), [HyCo2](#), [Spark](#), [ResAdapt](#), [AllocEmbed](#)).
- **Knowledge & Continual Adaptation**: acquiring, transferring, internalizing, and updating knowledge and capabilities across evolving tasks, while balancing plasticity, retention, and generalization ([TAGI](#), [AAG](#), [NesycD](#), [SKIntern](#), [SpaRTA](#), [SHIP](#), [DyPRAG](#)).
- **Agent Learning & Self-Improvement**: persistent memory, long-horizon decision-making, and learning from interaction and feedback to verifiably refine agent memory, policies, and learning procedures ([CITI](#), [KATE](#), [OPERATE](#), [Distill Yourself](#), [SLE](#), [BuPO](#), [Tool-RL-Box](#)).

👥 RESEARCH EXPERIENCE

Tencent (Project UP) Beijing

Jul. 2026 – Present

Pretraining Data and Models

- **Long-context pretraining data**: Built and delivered **186.8B tokens** across six 8K–512K length ranges and a separate **10B-token** training set; checked length distributions and answer annotations for training.
- **Data design and ablations**: Interleaved multiple QA turns through long documents to provide more useful training answers; compared ten data formats and mixtures. In early comparisons with the same target mixture, refining the QA structure raised peak MRCR content score across 8K–128K from **0.854 to 0.911**.
- **Sparse-attention evaluation**: Compared long-context quality and inference speed under matched settings. Identified bottlenecks in selecting context blocks and moving KV caches, showing why fewer selector operations alone do not guarantee faster inference.
- **Vision-code pretraining data**: Helped build SVG, sketch-to-HTML, and CAD datasets. Checked image-code consistency, removed visual duplicates, and filtered low-quality samples using visual embeddings and consistency scores; the SVG corpus contained **8.13M samples** (about **41B tokens**).

ByteDance Beijing

Jun. 2025 – Jun. 2026

Multimodal Understanding

- **AIGC detection**: Deployed efficient multimodal features for TikTok AIGC detection, improving online F1 by **>3%** and reducing compute by **30%–50%**; reused these features for embedding retrieval.
- **Efficient video reasoning**: Proposed [ResAdapt](#) and CAPO to adjust each frame's input resolution. Reduced visual tokens by **>90%**, processed up to **16×** more frames at the same compute, and achieved a **>15%** relative gain in long-video reasoning.
- **Efficient video retrieval**: Contributed to [AllocEmbed](#), which uses low-resolution previews to allocate frame resolution and retrieval-driven policy optimization (RDPO) to train the allocator. At matched visual-input budgets, LongRet Hit@1 improved by **5.69–9.09 percentage points** across four frozen backbones (four-subset averages); the allocator transferred without retraining.
- **Model efficiency**: Compared visual-token and KV-cache compression to measure their accuracy and compute costs; explored layer skipping and dynamic boundaries as further efficiency options.
- **Agent evaluation**: Built [OPERATE](#) to test when agents should gather information, act, or keep a plan as their environment changes. Scored decisions using execution records and comparisons of alternative actions under the same simulation seed.

- **Long-context compression:** Proposed [HyCo₂](#) to preserve a document's overall meaning and relevant evidence in a shorter input. Matched uncompressed performance with about **88.8%** fewer input tokens; achieved a **13.1% average relative gain** across seven knowledge-intensive QA tasks.
- **Survey and open source:** Core contributor to the [LCLM-Horizon survey](#), organizing research on data, models, training, and evaluation; curated and maintained [Awesome-LCLM](#) (**Curator & Maintainer**).

SELECTED PUBLICATIONS [FULL LIST]

Long-Context Data & Model Co-Design

[ACM TOIS'26] [Huanxuan Liao](#), Yao Xu, Shizhu He, Jun Zhao, Kang Liu. [HyCo₂: Hybrid Context Compression with Global Semantics and Local Evidence](#).

[arXiv'26] [Huanxuan Liao](#), Zhongtao Jiang, Yupu Hao, Yuqiao Tan, Shizhu He, Ben Wang, Jun Zhao, Kun Xu, Kang Liu. [ResAdapt: Adaptive Resolution for Efficient Multimodal Reasoning](#).

[arXiv'26] Song Jin, Zhongtao Jiang, Chenglei Shen, [Huanxuan Liao](#), Haozhe Chi, Zhiwei Wang, Kun Xu, Yong Liu. [Allocate Before You Embed: Adaptive Visual Input Allocation for Video Embeddings](#).

[AAAI'26] [Huanxuan Liao](#), Yixing Xu, Shizhu He, Guanchen Li, Xuanwu Yin, Dong Li, Emad Barsoum, Jun Zhao, Kang Liu. [Spark: Query-Aware Unstructured Sparsity with Recoverable KV Cache Channel Pruning](#).

[arXiv'25] Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, [Huanxuan Liao](#), Haoran Que, Zekun Wang, et al. [A Comprehensive Survey on Long Context Language Modeling](#). (**Core Contributor**)

Knowledge & Continual Adaptation

[ACL'26] [Huanxuan Liao](#), Shizhu He, Yupu Hao, Yequan Wang, Wenhao Teng, Xiangwen Liao, Jun Zhao, Kang Liu. [Spectral Disentanglement: Rank-Aware Task Adaptation for Rehearsal-free Continual Learning in LLMs](#).

[ACL'26] Wangtao Sun, Haotian Xu, [Huanxuan Liao](#), Xuanqing Yu, Zhongtao Jiang, Shizhu He, Jun Zhao, Kang Liu. [Shuttle Between Symbolic Instructions and Neural Parameters of Large Language Models](#).

[NeurIPS'26] Yuqiao Tan, Shizhu He, [Huanxuan Liao](#), Jun Zhao, Kang Liu. [Dynamic Parametric Retrieval Augmented Generation for Test-time Knowledge Enhancement](#). *TTCL Workshop*.

[AAAI'25] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Kang Liu, Jun Zhao. [Neural-Symbolic Collaborative Distillation: Advancing Small Language Models for Complex Reasoning Tasks](#).

[COLING'25] [Huanxuan Liao](#), Shizhu He, Yupu Hao, Xiang Li, Yuanzhe Zhang, Jun Zhao, Kang Liu. [SKIntern: Internalizing Symbolic Knowledge for Distilling Better CoT Capabilities into Small Language Models](#).

[COLING'25] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Shengping Liu, Kang Liu, Jun Zhao. [Awakening Augmented Generation: Learning to Awaken Internal Knowledge of Large Language Models for Question Answering](#).

[NeurIPS'24] [Huanxuan Liao](#), Shizhu He, Yao Xu, Yuanzhe Zhang, Yanchao Hao, Shengping Liu, Kang Liu, Jun Zhao. [From Instance Training to Instruction Learning: Task Adapters Generation from Instructions](#).

Agent Learning & Self-Improvement

[EMNLP'26] Yuqiao Tan, Minzheng Wang, Shizhu He, [Huanxuan Liao](#), Chengfeng Zhao, Qiunan Lu, Tian Liang, Jun Zhao, Kang Liu. [Bottom-up Policy Optimization: Your Language Model Policy Secretly Contains Internal Policies](#). **Oral**.

[ICML'26] Yupu Hao, Zhuoran Jin, [Huanxuan Liao](#), Kang Liu, Jun Zhao. [Why Multi-Step Tool-Use Reinforcement Learning Collapses and How Supervisory Signals Fix It](#). *FAGEN Workshop*.

♥ AWARDS AND SCHOLARSHIPS

National Scholarship ×3
Beijing Outstanding Graduate

2021 / 2022 / 2025
2023.06

🔗 PROFESSIONAL SERVICES

Conference Reviewing: ACL ARR 2024–2026. NeurIPS 2025–2026. ICML 2026. COLM 2026. ICLR 2025–2026. AAAI 2026.

Open Source: [Awesome-LCLM](#): Curator & Maintainer; [NVIDIA/kvpress](#): Contributor; [Distill Yourself](#): Core Logic & Code Contributor.